

# Within-session learning in DishBrain is likely explained by asymmetric feedback rather than by unpredictability

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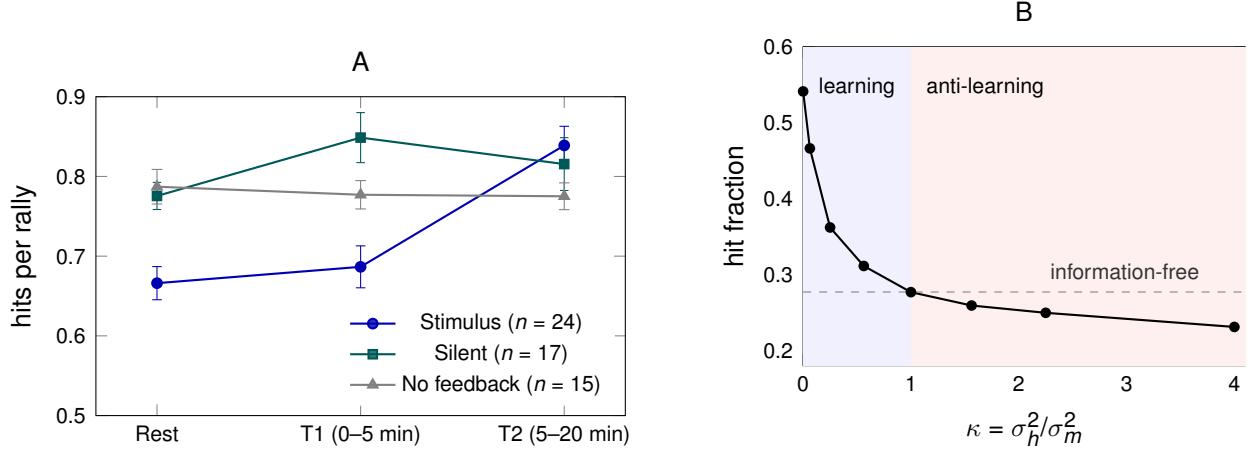
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Kagan et al.<sup>1</sup> showed that cortical cultures embedded in a closed-loop Pong game improve their play when a miss is followed by unpredictable stimulation and a hit by predictable stimulation, and interpreted this as the cultures acting to minimize surprise, as the free-energy principle proposes. The feedback after a miss, however, differs from the feedback after a hit not only in predictability. After a hit, all eight sensory electrodes are stimulated at 75 mV and 100 Hz for 100 ms; after a miss, random electrodes are stimulated at 150 mV, a voltage chosen to be “even more disruptive to the culture,”<sup>1</sup> for 4 s, and play resumes only after a further 4 s. The published design therefore cannot attribute the learning to unpredictability rather than to amplitude or to the eight-second time-out. The same feedback is used in the group’s recent comparison of cultures with reinforcement-learning agents,<sup>2</sup> and gameplay relative to Rest now serves as a read-out for drug effects,<sup>3</sup> so the question bears on current work as well. The closed-loop platform and the authors’ release of per-rally data make this question answerable. We address only this question and not the terminology of intelligence and sentience, which has been debated elsewhere.<sup>4,5</sup>

We re-analyzed the authors’ public gameplay data<sup>6</sup> with the culture as the unit<sup>7</sup> and hits per rally as the measure, comparing the first 5 minutes of gameplay (T1) with the remaining 15 (T2), as in the original study (Figure 1A; code at <https://github.com/kladkogex/20-watt-computer>, folder papers/dishbrain-scrambler/reanalysis; methods in Supplemental Information). Stimulus cultures improved within the session from 0.687 to 0.839 hits per rally (+0.152, Wilcoxon  $p = 0.0002$ ,  $n = 24$ ), with a positive mean gain in each of the three cell lines. Silent cultures, which receive no stimulation after a miss but the same eight-second withdrawal of input, did not improve ( $-0.033$ ,  $p = 0.35$ ,  $n = 17$ ), and neither did No feedback cultures ( $-0.002$ ,  $p = 1$ ,  $n = 15$ ). At T2 the three conditions did not differ (0.839, 0.815 and 0.775; Kruskal–Wallis  $p = 0.22$ ,  $n = 56$ ): Stimulus reached, within the session, the level that the other two groups had from the start. The published comparisons express gameplay relative to each culture’s Rest session, but Rest baselines differed between the groups (0.666, 0.775 and 0.787 hits per rally; Kruskal–Wallis  $p = 7 \times 10^{-5}$ ), with the Stimulus cultures lowest. The groups also differed in composition: Silent contains no primary mouse cultures, and No feedback contains 2 human GFP-iPSC cultures against 12 in Stimulus. The data thus show within-session improvement under the complete miss feedback only, and no condition ending above the others.

A simple mechanism accounts for this without prediction. Suppose each miss moves a network’s setting  $\psi$  by a random, symmetric step of variance  $\sigma_m^2$ , and each hit by one of variance  $\sigma_h^2$ . For small steps the setting diffuses with the state-dependent coefficient  $D = \frac{1}{2}\sigma_m^2[\kappa + (1 - \kappa)P(\psi)]$ , where  $P$  is the miss probability and  $\kappa = \sigma_h^2/\sigma_m^2$ , and the stationary density is  $\rho \propto 1/[\kappa + (1 - \kappa)P]$  (derivation in Supplemental Information).<sup>8</sup> Settings that miss less are kept longer, so the long-run miss rate falls below that of a random setting if and only if  $\kappa < 1$ , and rises above it for  $\kappa > 1$  (Figure 1B). No step is directed; good settings are not sought, only left alone, as in the stimulus regulation principle proposed for cultures<sup>9</sup> and learning by stimulation avoidance.<sup>10</sup> Because  $\sigma_m^2$  multiplies  $D$ ,



**Figure 1.** Re-analysis of the published data and the perturbation account. (A) Hits per rally at Rest and in the first 5 (T1) and last 15 minutes (T2) of gameplay in the three feedback conditions of Kagan et al.,<sup>1,6</sup> mean  $\pm$  SEM over cultures. Only Stimulus improves within the session ( $p = 0.0002$ ); at T2 the conditions do not differ ( $p = 0.22$ ); Rest baselines differ ( $p = 7 \times 10^{-5}$ ). (B) A closed-loop Pong model whose only teacher is a random kick to its sensorimotor weights after each approach, of variance  $\sigma_m^2$  after a miss and  $\sigma_h^2$  after a hit: hit fraction in the last three quarters of a session against  $\kappa = \sigma_h^2 / \sigma_m^2$  (400 simulated cultures per point). Performance crosses the information-free level (dashed) at  $\kappa = 1$ ; the model computes no prediction, reward or error.

the size of the perturbation sets how fast a network learns but not where it ends, which is what the re-analysis shows: a fast gain under the strongest feedback and equal end levels. This account is not opposed to a free-energy description, and a surprise-minimizing system may implement exactly this stabilization, but it is the simpler mechanism the published data support, and the test below separates the two.

The account does not explain everything. In a spiking model of the culture on the chip's geometry, calibrated to the Rest recordings, two local plasticity rules without reward give  $\kappa \approx 0.6$ – $0.8$  per approach.<sup>8</sup> This contingency comes from the eight-second time-out rather than from the 150 mV pulses, so the model predicts learning in Silent as well, which the data do not show. Either real 150 mV stimulation perturbs synapses more than the model's local rules do, or Silent's lack of improvement reflects its high starting level (0.849 hits per rally at T1). Stimulation is also unlikely to perturb synapses isotropically: a 100 Hz burst can itself potentiate specific pathways (Supplemental Information).

One experiment decides. In protocol F, a hit is followed by the same eight-second time-out as a miss, and only a miss is followed by 150 mV random stimulation. If unpredictability teaches, F learns as the published protocol does. Under the perturbation account, F learns only if the amplitude of the stimulation, not the time-out, supplies the contingency. Two analyses of existing recordings would also discriminate. First, if a miss moves the network's state, for example the evoked response to a fixed probe, further than a hit does, then across cultures the within-session gain should grow with that difference. Second, in Silent, the number of spontaneous bursts after each miss should predict a culture's gain.

The two accounts agree on every published condition and differ on protocol F. We offer it as a proposal for an experiment that the authors' platform is well equipped to run.

## Declaration of interests

The author declares no competing interests.

## Supplemental information

Document S1. Methods of the re-analysis and statistics, derivation of the stationary density, and the closed-loop model.

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